Leveraging Large Language Models for Adverse Drug Event Detection: A Comparative Study of Token and Span-Based Named Entity Recognition

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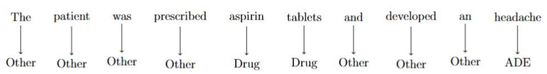
INTRODUCTION

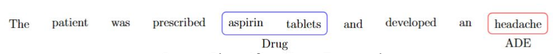
Adverse Drug Events (ADEs) pose a critical risk to patient safety, with over **2.15 million reported cases** and **163,000 deaths in 2023**. Clinical notes provide valuable information for ADE detection, but their unstructured nature makes automated extraction challenging. Large language models (LLMs) offer new opportunities for improving ADE detection through named entity recognition (NER) and relationship extraction (RE). In this work, **we compare token-based and span-based NER approaches**, and explore ensemble strategies to improve **end-to-end ADE detection using the 2018 n2c2 dataset**.

METHODS

We used 505 discharge summaries from the 2018 n2c2 challenge, annotated for Drugs, ADEs, and related attributes. Notes were split into sentences using NLTK, and entities were grouped into three classes: Drug, ADE, and Other. Four LLMs were selected to represent a range of architectures and release years: Qwen2.5 (0.5B), RoBERTa-large, ModernBERT, and BERT-base. We evaluated three tasks: token classification, span classification, and relationship extraction (RE). Token classification treated NER as sequence labeling, while span classification used the PL-Marker (SpanMarker) framework. RE was framed as binary classification of Drug–ADE links, evaluated under both golden label and end-to-end settings. Finally, we tested ensemble methods including majority voting, probability-based aggregation, and meta-learners such as XGBoost and MLPs.



Token Classification Example



Span Classification Example

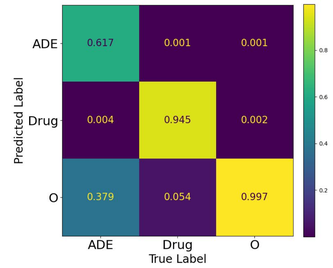
Dataset	avg len	ADE	Drug	Other
train	383	1665	18039	96350
test	434	1057	11905	74765

Table 1: Dataset Metrics

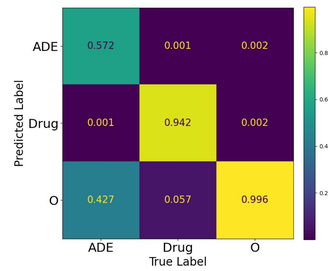
RESULTS

Model Name	Exp. Type	ADE-F1	Drug-F1	Macro-F1
ModernBERT	token	0.49258	0.94026	0.8099
RoBERTa-large	token	**0.54813**	**0.94612**	**0.83043**
BERT-base-cased	token	0.46102	0.93781	0.79849
Qwen2.5	token	0.21021	0.79887	0.66714
BERT-base-cased	span	0.42832	0.93247	0.78575
RoBERTa-large	span	**0.4774**	**0.94052**	**0.80488**
Majority vote	ensemble	**0.50898**	0.94579	**0.8173**
Minority vote	ensemble	0.32744	0.76664	0.69496
Highest proba vote	ensemble	0.33554	0.93311	0.75509
Sum proba vote	ensemble	0.42004	0.94584	0.78765
Meta vote XGB	ensemble	0.50234	0.94013	0.81312
Meta proba LGBM	ensemble	0.44869	**0.94599**	0.79724
Meta vote + proba MLP	ensemble	0.37536	0.94166	0.77131

Table 2: Weighted Metrics on Zero Shot Classification



Roberta Token CM



Roberta Span CM

Model	Accuracy	Macro-Pr	Macro-Re	Macro-F1
ModernBERT	0.7607	0.7185	0.675	0.6875
Qwen2.5	0.7451	0.6981	0.7005	0.6992
RoBERTa-large	0.7671	0.7249	0.693	0.704
BERT-base-cased	**0.7814**	**0.7405**	**0.741**	**0.7407**

Table 3: Golden Label Relationship Extraction

Model / Method	Exp. Type	Accuracy	Macro-Pr	Macro-Re	Macro-F1
ModernBERT	token	0.247	0.2304	0.2253	0.2277
RoBERTa-large	token	0.2876	0.2715	0.2439	0.2542
RoBERTa-large	span	0.2751	0.2556	0.2413	0.2474
BERT-base-cased	token	**0.3213**	**0.2766**	**0.2762**	**0.2764**
BERT-base-cased	span	**0.3005**	**0.2564**	**0.2718**	**0.263**
Qwen2.5	token	0.1601	0.1309	0.2119	0.1469
Majority Vote	Ensemble	**0.3262**	**0.2808**	**0.2901**	**0.285**
XGB Vote	Ensemble	0.3199	0.275	0.2868	0.2802
MLP Vote and Proba	Ensemble	0.1175	0.0865	0.1339	0.1051

Table 4: End-to-End Relation Classification Metrics

RESULTS

RoBERTa-large (token classification) achieved the **best macro-F1 (0.83), excelling in both Drug and ADE recognition**. Token models generally outperformed span models, especially for ADE recall, though span models remained useful for boundary-sensitive Drug identification. **Ensembles further boosted performance**: majority voting and XGBoost reduced precision–recall gaps for ADEs, while minority voting occasionally caught difficult ADEs with low consensus. **For relation extraction, BERT-base performed best with golden labels (F1 = 0.74)**. End-to-end RE was harder, but **ensembles improved stability by mitigating error propagation**.

CONCLUSION

ADE detection remains challenging due to low recall on rare entities, whereas **Drug recognition is more reliable**. Performance depended more on architecture and task alignment than model size or recency. Token-level models offered stronger ADE recognition, while span models were useful for boundary precision. Ensembles proved effective in end-to-end RE by leveraging model diversity. Future work should address class imbalance, explore multi-task learning, and incorporate biomedical pretraining to strengthen ADE detection pipelines.

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